

Exercise I.

[25 marks]

Compute the following integrals:

1. $\int \frac{1}{\sqrt{1-x^2} \arcsin x} dx,$

4. $\int \frac{3x-4}{(x-1)(x-2)} dx,$

2. $\int \frac{1}{\cos^4 x} dx,$

5. $\int \frac{3x^3 - 5x^2 - 3x + 4}{x^2 - 3x + 2} dx.$

3. $\int x \frac{\sin x}{\cos^5 x} dx,$

Exercise II.

[15 marks]

Let $I_n = \int_0^{\frac{1}{2}} \frac{x^n}{x-1} dx$ for all $n \in \mathbb{N}$.

1. Calculate I_0 .

2. Show that $\lim_{n \rightarrow \infty} I_n = 0$.

3. Show that $I_{n+1} - I_n = \frac{1}{(n+1)2^{n+1}}$.

4. Deduce the value of $\sum_{n=0}^{\infty} \frac{1}{(n+1)2^{n+1}}$.

Exercise III.

[20marks]

1. Calculate the following limit: $\lim_{n \rightarrow \infty} \sqrt[n]{\prod_{k=1}^n \frac{n+k}{n}}.$

2. Let $g(x) = \int_x^{x-1} \ln(|\ln t|) dt$. Compute $g'(x)$ and specify the domain of differentiability of g .

Exercise IV.

[15 marks]

Study the natures of the following series:

$$\sum_{n=1}^{\infty} \left(\frac{n}{n+1} \right)^{n^2}, \quad \sum_{n=2}^{\infty} (-1)^n \frac{1}{\ln n}, \quad \sum_{n=2}^{\infty} \frac{n \arctan n}{(n+1) \ln n}.$$

Exercise V.

[25 marks]

1. Study the natures of $I = \int_0^{+\infty} \frac{\sqrt{x-x}}{x+x^3} dx$ and $J = \int_1^3 \frac{1}{\sqrt{(3-x) \ln x}} dx.$

2. Let $\alpha > 0$.

(a) Show that $K = \int_e^{+\infty} \frac{\cos x}{x^\alpha} dx$ is convergent.

(b) Show that $L = \int_e^{+\infty} \ln \left(1 + \frac{\cos x}{x^\alpha} \right) dx$ is convergent if $\alpha > \frac{1}{2}$ and divergent if $\alpha \leq \frac{1}{2}$.

(c) Deduce the nature of $N = \int_e^{+\infty} \frac{\cos x}{x^{\frac{1}{2}}} dx.$

Recall that; $\frac{o}{o}(g(x)) = \varepsilon(x)g(x)$ in a neighborhood of ∞ with $\varepsilon(x) \rightarrow 0$ as $x \rightarrow \infty$.



Exercise I. Compute the following integrals (5x5 pts)

1. $\int \frac{1}{\sqrt{1-x^2} \arcsin x} dx = \int \frac{d(\arcsin x)}{\arcsin x} = \ln |\arcsin x| + C.$

2. $\int \frac{1}{\cos^4 x} dx = \int \frac{1}{\cos^2 x} \cdot \frac{1}{\cos^2 x} dx = \int (1 + \tan^2 x) d(\tan x) = \tan x + \frac{\tan^3 x}{3} + C.$

3. $\int x \frac{\sin x}{\cos^5 x} dx = x \cdot \frac{1}{4 \cos^4 x} - \frac{1}{4} \int \frac{1}{\cos^4 x} dx = \frac{x}{4 \cos^4 x} - \frac{\tan x}{4} - \frac{\tan^3 x}{12} + C.$

4. $\int \frac{3x-4}{(x-1)(x-2)} dx = \int \frac{1}{x-1} dx + \int \frac{2}{x-2} dx = \ln|x-1| + 2 \ln|x-2| + C$

5. $\int \frac{3x^3 - 5x^2 - 3x + 4}{x^2 - 3x + 2} dx = \int \left[3x + 4 + \frac{3x-4}{x^2-3x+2} \right] dx$
 $= \frac{3x^2}{2} + 4x + \int \frac{3x-4}{(x-1)(x-2)} dx$
 $= \frac{3}{2}x^2 + 4x + \ln|x-1| + 2 \ln|x-2| + C.$

Partial fractions:
 $u = x, v' = \frac{\sin x}{\cos^5 x}$
 $w = 1 \Rightarrow v = \int \frac{d(\cos x)}{\cos^5 x} = \frac{1}{4 \cos^4 x}$

Partial fractions for (4):
 $\frac{3x-4}{(x-1)(x-2)} = \frac{A}{x-1} + \frac{B}{x-2}$
 $\Rightarrow 3x-4 = A(x-2) + B(x-1)$
 $\cdot x=1 \Rightarrow A=1 \text{ and } x=2 \Rightarrow B=2$
 $\frac{3x^3-5x^2-3x+4}{x^2-3x+2} = 3x + 4 + \frac{3x-4}{x^2-3x+2}$
 $= 3x + 4 + \frac{3x^2-9x^2+6x+4}{x^2-3x+2}$
 $= 3x + 4 + \frac{4x^2-9x+4}{x^2-3x+2}$
 $= 3x + 4 + \frac{4x^2-12x+8-3x+4}{x^2-3x+2}$
 $= 3x + 4 + \frac{4x^2-12x+12}{x^2-3x+2}$
 $= 3x + 4 + \frac{4(x^2-3x+3)}{x^2-3x+2}$
 $= 3x + 4 + \frac{4(x^2-3x+2+1)}{x^2-3x+2}$
 $= 3x + 4 + \frac{4(x^2-3x+2)}{x^2-3x+2} + \frac{4}{x^2-3x+2}$
 $= 3x + 4 + 4 + \frac{4}{x^2-3x+2}$
 $= 3x + 8 + \frac{4}{x^2-3x+2}$

Exercise II. Let $I_n = \int_0^{1/2} \frac{x^n}{x-1} dx$. (15 pts = 2+5+4+4 pts)

1. calculate I_0 .

$I_0 = \int_0^{1/2} \frac{1}{x-1} dx = \ln|x-1| \Big|_0^{1/2} = \ln \frac{1}{2} = -\ln 2.$

2. Show that $\lim_{n \rightarrow \infty} I_n = 0$.

$\lim_{n \rightarrow \infty} I_n = \lim_{n \rightarrow \infty} \int_0^{1/2} x^n \cdot \frac{1}{x-1} dx \stackrel{MVT}{=} \lim_{n \rightarrow \infty} c^n \int_0^{1/2} \frac{1}{x-1} dx = 0.$

$|I_n| = \left| \int_0^{1/2} \frac{x^n}{x-1} dx \right| \leq \int_0^{1/2} \left| \frac{x^n}{x-1} \right| dx \leq \int_0^{1/2} \frac{x^n}{1-x} dx \leq \left(\frac{1}{2}\right)^n \int_0^{1/2} \frac{1}{1-x} dx \xrightarrow{n \rightarrow \infty} 0.$ By Sandwich Rule, $I_n \rightarrow 0$.

$0 \leq |I_n| \leq \int_0^{1/2} \frac{x^n}{1-x} dx \leq \frac{(1/2)^n}{1/2} \cdot \int_0^{1/2} 1 dx \leq (1/2)^n \cdot \frac{1}{2} \xrightarrow{n \rightarrow \infty} 0.$ By S. rule, $I_n \rightarrow 0$.

3. Show that $I_{n+1} - I_n = \frac{1}{(n+1)2^{n+1}}$.

$I_{n+1} - I_n = \int_0^{1/2} \frac{x^{n+1} - x^n}{x-1} dx = \int_0^{1/2} \frac{x^n(x-1)}{(x-1)} dx = \int_0^{1/2} x^n dx = \frac{x^{n+1}}{n+1} \Big|_0^{1/2} = \frac{1}{(n+1)2^{n+1}}.$

4. Deduce the value of $\sum_{n=0}^{\infty} \frac{1}{(n+1)2^{n+1}}$.

$\sum_{n=0}^{\infty} \frac{1}{(n+1)2^{n+1}} = \sum_{n=0}^{\infty} (I_{n+1} - I_n) = \lim_{n \rightarrow \infty} (I_{n+1} - I_0) = 0 - (-\ln 2) = \ln 2.$

Exercise III. (10+10 pts)

1. Calculate the following limit:

$\lim_{n \rightarrow \infty} \left(\prod_{k=1}^n \frac{n+k}{n} \right)^{1/n} = \lim_{n \rightarrow \infty} e^{\frac{1}{n} \sum_{k=1}^n \ln \left(\frac{n+k}{n} \right)} = \lim_{n \rightarrow \infty} e^{\frac{1}{n} \sum_{k=1}^n \ln \left(1 + \frac{k}{n} \right)}$
 $= e^{\int_0^1 \ln(1+x) dx}$
 $= e^{[x \ln(1+x) - \int_0^1 \frac{x}{1+x} dx]} = e^{[x \ln(1+x) - (x - \ln(1+x))]_0^1} = e^{(\ln 2 - 1 + \ln 2) - (1 - 0)} = e^{2 \ln 2 - 1} = \frac{4}{e}.$

2. Let $g(x) = \int_1^x \ln(\ln t) dt$.

Compute $g'(x)$: $g'(x) = (x-1)' \ln(\ln(x-1)) - (x)' \ln(\ln x) = \ln(\ln(x-1)) - \ln(\ln x).$



Specify the domain of differentiability of g :

* $f(t) = \ln(\ln|t|)$ is continuous on $]-\infty, -1[\cup]1, \infty[$. (1pt)

* $u(x) = x^{-1}$ & $v(x) = x$ are diff on $\mathbb{I}_1 = \mathbb{R}$. (1pt)

* $u(x), v(x) \in]-\infty, -1[\Leftrightarrow x^{-1} < -1$ and $x < -1 \Leftrightarrow x < 0$ and $x < -1 \Leftrightarrow x < -1 \Leftrightarrow x \in]-\infty, -1[= \mathbb{I}_2$ (1pt)

* $u(x), v(x) \in]1, \infty[\Leftrightarrow x^{-1} > 1$ and $x > 1 \Leftrightarrow x > 2$ and $x > 1 \Leftrightarrow x > 2 \Leftrightarrow x \in]2, \infty[= \mathbb{I}_3$. (1pt)

So the domain of diff of g is $J = (\mathbb{I}_1 \cap \mathbb{I}_2) \cup (\mathbb{I}_1 \cap \mathbb{I}_3) = \mathbb{I}_2 \cup \mathbb{I}_3 \subset]-\infty, -1[\cup]2, \infty[$. (1pt)

Exercise IV. Study the nature of the following series: (5x3pts)

• $\sum_{n=1}^{\infty} \left(\frac{n}{n+1}\right)^{n^2}$; $\lim_{n \rightarrow \infty} |a_n|^{1/n} = \lim_{n \rightarrow \infty} \left(\frac{n}{n+1}\right)^n = \lim_{n \rightarrow \infty} \frac{1}{\left(\frac{n+1}{n}\right)^n} = \lim_{n \rightarrow \infty} \frac{1}{\left(1+\frac{1}{n}\right)^n} = \frac{1}{e} < 1$. (2pts) (1pt)

By Cauchy's Root test, the series converges absolutely. (1pt)

• $\sum_{n=2}^{\infty} (-1)^n \frac{1}{\ln n}$ b_n

• $b_n > 0$ (for n large) (1pt)

• $\{b_n\} \downarrow$ ($\because \ln(n+1) > \ln(n) > 0 \Rightarrow b_{n+1} = \frac{1}{\ln(n+1)} < b_n = \frac{1}{\ln n}$) (1pt)

• $\lim_{n \rightarrow \infty} b_n = 0$ (1pt)

By Leibniz's alternating series test, the series converges. (1pt)

• $\sum_{n=2}^{\infty} \frac{n \arctan n}{(n+1) \ln n}$

1st way: $\frac{n \arctan n}{(n+1) \ln n} \sim \frac{n \cdot \pi/2}{n \ln n} = \pi/2 \cdot \frac{1}{\ln n} > 0$ for n -large. (2pts) (1pt)

$\sum_{n=2}^{\infty} \frac{1}{\ln n} = \sum_{n=2}^{\infty} \frac{1}{n^0 \ln n}$ divergent (Bertrand ex), hence the series is divergent. (1pt) (equiv test)

2nd way: $\frac{n \arctan n}{(n+1) \ln n} \geq \frac{n \arctan 2}{2n^2} = \frac{\arctan 2}{2} \cdot \frac{1}{n} > 0$ (2pts) (1pt)

$\sum \frac{1}{n}$ divergent (Riemann ex), hence the series is divergent (comparison test) (1pt)

3rd way: $n^{\alpha} \cdot \frac{n \arctan n}{(n+1) \ln n} \sim \frac{n^2 \pi/2}{n \ln n} = \pi/2 \cdot \frac{n}{\ln n} \rightarrow \infty$ and $\alpha = 1 \leq 1$, then the series div. (1pt) (1pt) (1pt)

Exercise V. (25 pts)

1. Study the natures of $I = \int_0^{\infty} \frac{\sqrt{x-x^2}}{x+x^3} dx$ and $J = \int_1^3 \frac{1}{\sqrt{(3-x)\ln n}} dx$.

For I : near 0^+ , $f(x) \sim \frac{1}{\sqrt{x}} > 0$. $\int_0^1 \frac{1}{\sqrt{x}} dx$ conv (Riemann) $\Rightarrow \int_0^1 f(x) dx$ conv. (1/2pt)
near ∞ , $f(x) \sim -\frac{1}{x^2} < 0$. $\int_1^{\infty} \frac{1}{x^2} dx$ conv (Riemann) $\Rightarrow \int_1^{\infty} f(x) dx$ conv. (1/2pt) $\Rightarrow I$ is conv.

For J : near 1^+ , $g(x) = \frac{1}{\sqrt{(3-x)\ln(1+(n-1))}} \sim \frac{1}{\sqrt{2} (n-1)^{1/2}} > 0$. $\int_1^2 \frac{1}{(n-1)^{1/2}} dx$ conv (Riemann) $\Rightarrow \int_1^2 g(x) dx$ conv. (1/2pt)
near 3^- , $g(x) \sim \frac{1}{\sqrt{\ln 3} (3-x)^{1/2}} > 0$. $\int_2^3 \frac{1}{(3-x)^{1/2}} dx$ conv $\Rightarrow \int_2^3 g(x) dx$ conv. (1/2pt) $\Rightarrow J$ is conv.

2. let $\alpha > 0$.

a) S.T $K = \int_e^{\infty} \frac{\cos x}{x^{\alpha}} dx$ is conv; $K = \int_e^{\infty} \frac{1}{x^{\alpha}} \cos x = \frac{\sin x}{x^{\alpha}} \Big|_e^{\infty} + \alpha \int_e^{\infty} \frac{\sin x}{x^{\alpha+1}} dx = [0 - \frac{\sin e}{e^{\alpha}}] + \alpha K_1$ (2pts) (1pt) link now

K_1 is absolutely conv (By comparison $|\frac{\sin x}{x^{\alpha+1}}| < \frac{1}{x^{\alpha+1}}$; $\alpha+1 > 1$), hence K is convergent. (1pt)

b) S.T $L = \int_e^{\infty} \ln(1 + \frac{\cos x}{x^{\alpha}}) dx$ is conv if $\alpha > 1/2$ and div if $\alpha \leq 1/2$.

$\ln(1 + \frac{\cos x}{x^{\alpha}}) = \frac{\cos x}{x^{\alpha}} - \frac{\cos^2 x}{2x^{2\alpha}} + o(\frac{\cos^2 x}{x^{2\alpha}})$. (1pt)

For $\alpha > 1/2$: $\int_e^{\infty} \frac{\cos x}{x^{\alpha}} dx$ is conv (part a, $\alpha > 0$) (1pt)
 $\int_e^{\infty} \frac{\cos^2 x}{x^{2\alpha}} dx$ is conv ($0 \leq \frac{\cos^2 x}{x^{2\alpha}} \leq \frac{1}{x^{2\alpha}}$; $2\alpha > 1$) By Comp test & Riemann (2pts)
 $\int_e^{\infty} o(\frac{\cos^2 x}{x^{2\alpha}}) dx$ is conv ($\because \int_e^{\infty} \frac{\cos^2 x}{x^{2\alpha}} dx$ is absol conv) (1pt) $\Rightarrow L$ is conv for $\alpha > 1/2$



For $\alpha \leq 1/2$: $\ln(1 + \frac{\cos n}{x^\alpha}) - \frac{\cos n}{x^\alpha} \sim_{\infty} -\frac{1}{2} \frac{\cos^2 n}{x^{2\alpha}} < 0$ (near ∞). (1pt)

Since $\int_e^\infty \frac{\cos^2 n}{x^{2\alpha}} dn = \frac{1}{2} \int_e^\infty \frac{1 + \cos(2n)}{x^{2\alpha}} dn$ div ($\because \int_e^\infty \frac{1}{x^{2\alpha}} dn$ div ($2\alpha \leq 1$) & $\int_e^\infty \frac{\cos 2n}{x^{2\alpha}} dn$ conv)

Then by equivalent test $\int_e^\infty (\ln(1 + \frac{\cos n}{x^\alpha}) - \frac{\cos n}{x^\alpha}) dn$ is div. (1pt)

Now, $\left\{ \begin{array}{l} \int_e^\infty (\ln(1 + \frac{\cos n}{x^\alpha}) - \frac{\cos n}{x^\alpha}) dn \text{ is div} \\ \int_e^\infty \frac{\cos n}{x^\alpha} dn \text{ conv} \end{array} \right\} \xRightarrow{\text{sum}} \int_e^\infty \ln(1 + \frac{\cos n}{x^\alpha}) dn \text{ is div.}$ (1/2pt)

c) Deduce the nature of $N = \int_e^\infty o(\frac{\cos n}{x^{3/4}}) dn$.

$$\ln(1 + \frac{\cos n}{x^{3/4}}) = \frac{\cos n}{x^{3/4}} + o(\frac{\cos n}{x^{3/4}}) \Rightarrow o(\frac{\cos n}{x^{3/4}}) = \ln(1 + \frac{\cos n}{x^{3/4}}) - \frac{\cos n}{x^{3/4}}. \quad \text{1pt}$$

Now, $\left\{ \begin{array}{l} \int_e^\infty \ln(1 + \frac{\cos n}{x^{3/4}}) dn \text{ conv } (\alpha = 3/4 > 1/2) \\ \int_e^\infty \frac{\cos n}{x^{3/4}} dn \text{ conv } (\alpha = 3/4 > 0) \end{array} \right\} \Rightarrow N = \int_e^\infty o(\frac{\cos n}{x^{3/4}}) dn \text{ is conv.}$ (1/2pt)

Note that $\frac{\cos x}{x^{1/4}} \cdot \frac{\cos x}{x^{3/4}} = \frac{\cos^2 x}{x} = o(\frac{\cos x}{x^{3/4}}) \Rightarrow \int_e^\infty o(\frac{\cos x}{x^{3/4}}) dx$ is div +2 1/2 Bonus
(do $\int_e^\infty |\frac{\cos x}{x^{3/4}}| dx$ is not conv.)

Conclusion: Since $\int_e^\infty \frac{\cos n}{x^{3/4}} dn$ is not absol conv, then $\int_e^\infty o(\frac{\cos n}{x^{3/4}}) dn$ is either convergent or divergent (depends on $\varepsilon(n)$).